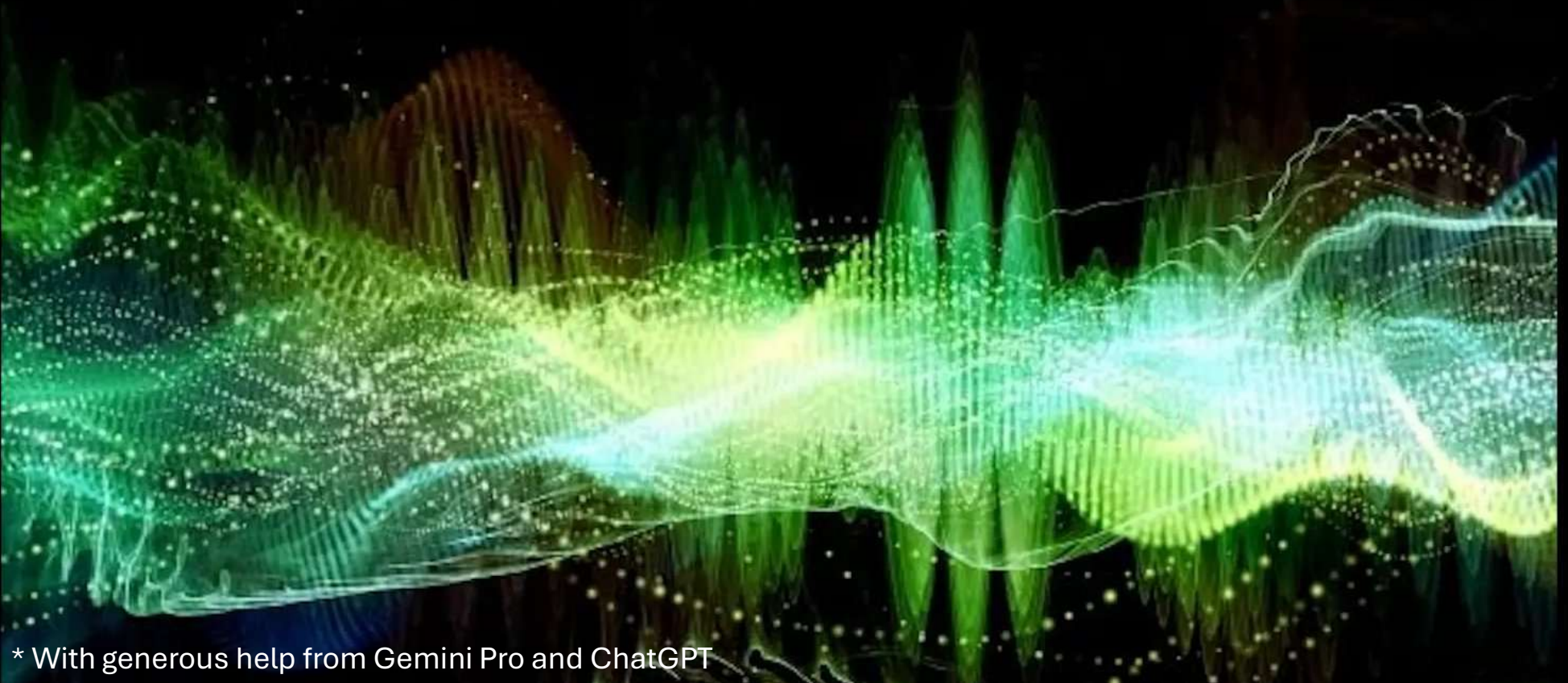


ECEN 4/5632 Theory and Application of Digital Filters

Lecture 2, 8-25-26, 8-27-26

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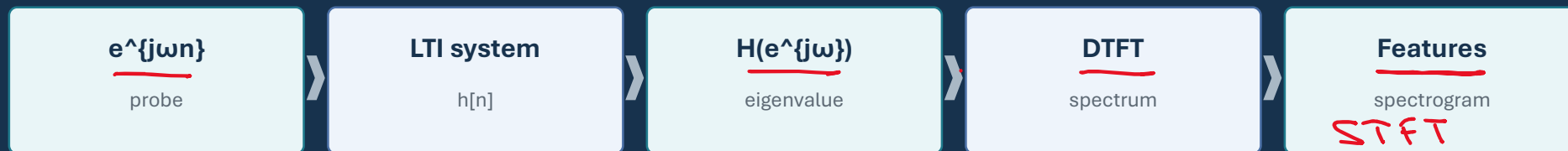


* With generous help from Gemini Pro and ChatGPT

Eigenfunctions and the DTFT

Week 1 • Lecture 2 • 75 minutes

Why sinusoids survive LTI systems—and why frequency-domain multiplication is so powerful



Today's learning objectives

00-10 MIN

Move from time-domain mechanics to frequency-domain structure.

- **Prove that complex exponentials are eigenfunctions of LTI systems.**
- Define the DTFT and inverse DTFT and explain 2π periodicity.
- Use time shift, convolution, and Parseval properties.
- Derive frequency response directly from a linear constant-coefficient difference equation.
- Interpret a one-pole filter in time and frequency domains.

Graduate expectation

Be able to state convergence or stability assumptions when a transform/frequency response is used.

Recap: an LTI system is convolution by $h[n]$

00-10 MIN

Now choose an input that makes convolution simplify dramatically.

$$y[n] = \sum_k h[k] x[n - k]$$

$$x[n] = e^{j\omega n}$$

$$\begin{aligned} y[n] &= \sum_k h[k] e^{j\omega(n-k)} \\ &= e^{j\omega n} \sum_k h[k] e^{-j\omega k} \end{aligned}$$

Notice

The input waveform $e^{j\omega n}$ factors back out. The system can only multiply it by a complex number.

$$x[n] = \underline{e^{j\omega n}} = \cos \omega n + j \sin \omega n$$

$$y[n] = \sum_k h[k] e^{j\omega(n-k)} = \underline{e^{j\omega n}} \sum_k h[k] e^{-j\omega k}$$

fixed for specific ω
eigenfunction
constant wrt n
 $\triangleq H(e^{j\omega})$

linear algebra

$$A \underline{v} = \lambda \underline{v}$$

eigenvectors

eigenvalue
 $\triangleq H(e^{j\omega})$

Complex exponentials are eigenfunctions of every LTI system

00-10 MIN

The eigenvalue depends on frequency.

$$y[n] = H(e^{j\omega})e^{j\omega n}$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k}$$

What stays the same

- Frequency ω 
- Exponential form 
- Infinite-duration mode 

What changes

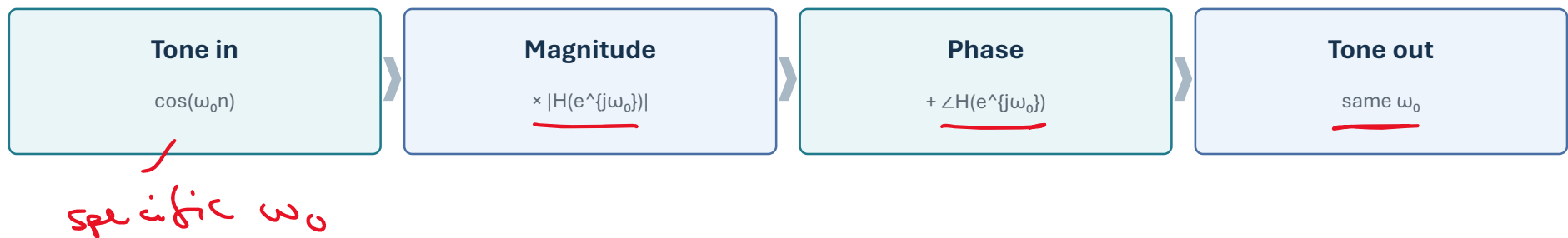
- Magnitude: $|H(e^{j\omega})|$ 
- Phase: $\angle H(e^{j\omega})$ 
- Possibly complete cancellation

Frequency response is a gain-and-phase lookup table

10-35 MIN

One complex number for each normalized frequency ω .

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$$



Audio EQ

A graphic equalizer is approximately a deliberately shaped $|H(e^{j\omega})|$.

The DTFT analyzes a sequence as a continuous function of frequency

10-35 MIN

Time is discrete; frequency remains continuous.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \quad x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Forward DTFT

Analysis

- Sequence $x[n] \rightarrow$ function $X(e^{j\omega})$
- Weighted sum of rotating phasors
- Requires convergence in an appropriate sense

Inverse DTFT

Synthesis

- Integrates one period of spectrum
- Reconstructs each integer sample
- Any interval of length 2π works

$$X(e^{j\omega}) = \sum_n x[n] e^{-j\omega n}$$

$$\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega = x[n]$$

contiguous interval of length 2π

DTFT $e^{j\omega n}$ are the basis functions

inverse DTFT

DTFT is the dual of the Fourier Series

Why is the DTFT 2π -periodic?

10-35 MIN

Because sampled-time exponentials cannot distinguish ω from $\omega + 2\pi k$. *integers*

$$e^{-j(\omega + 2\pi)n} = e^{-j\omega n} e^{-j2\pi n} = e^{-j\omega n}$$

$$X(e^{j(\omega + 2\pi)}) = X(e^{j\omega})$$

worth pointing out
periodicity of ω is a
consequence of having
DT sequences



same spectral content repeats every 2π

$$e^{-j(\omega + 2\pi)n} = e^{-j\omega n} \underbrace{e^{-j2\pi n}}_{\text{complicated way of saying 1}} = e^{-j\omega n} \Rightarrow X(e^{j(\omega + 2\pi)}) = X(e^{j\omega})$$

Normalized frequency $\leftrightarrow$ physical frequency

10-35 MIN

Week 2 will make this relationship central.

f_s : sampling frequ

$$\omega = 2\pi \frac{f}{f_s} \iff f = \frac{\omega}{2\pi} f_s$$

Example

At $f_s=48$ kHz: $\omega=\pi/2 \leftrightarrow 12$ kHz, and $\omega=\pi \leftrightarrow 24$ kHz.

$$\omega = \pi \iff f = \frac{f_s}{2}$$

- The interval $-\pi \leq \omega < \pi$ represents one unique frequency period.
- Frequencies outside that interval wrap into it modulo 2π .

normalized freq = $\frac{f \text{ in cycles/sec or Hz}}{f_s \text{ in samples/sec}}$

= $\frac{f}{f_s}$ [cycles/sample]

$\Rightarrow \omega = 2\pi \frac{f}{f_s}$ rad/sample

$-\pi \leq \omega < \pi$

is one unique frequency period

normalized freq ω

Two DTFT pairs worth knowing immediately

10-35 MIN

Use them as anchors—not as a substitute for reasoning.

$$\delta[n] \longleftrightarrow 1$$

$$a^n u[n] \longleftrightarrow \frac{1}{1 - ae^{-j\omega}}, \quad |a| < 1$$

$$\delta[n - n_0] \longleftrightarrow e^{-j\omega n_0}$$

Why the condition?

For $|a| < 1$, the geometric series is absolutely summable; $|a| \geq 1$ does not converge on the unit circle in the ordinary sense.

$$\delta[n]: \sum_n \underbrace{\delta[n] e^{-j\omega n}}_{= \delta[n] e^{-j\omega 0}} = \sum_n \delta[n] = 1$$

$$\delta[n - n_0]: \sum_n \underbrace{\delta[n - n_0] e^{-j\omega n}}_{= \delta[n - n_0] \cdot e^{-j\omega n_0}} = e^{-j\omega n_0}$$

DFT of $a^n \cdot u[n]$

$$\sum_{n=0}^{\infty} a^n u[n] e^{-j\omega n} = \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (a e^{-j\omega})^n = \frac{1}{1 - a e^{-j\omega}}$$

$|a e^{-j\omega}| < 1$ for convergence

$$|a e^{-j\omega}| = |a| < 1$$

Time shifting becomes linear phase

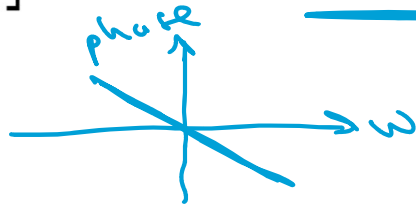
35-55 MIN

Delays are easy to recognize in the frequency domain.

$$x[n - n_0] \longleftrightarrow \underline{e^{-j\omega n_0}} X(e^{j\omega})$$

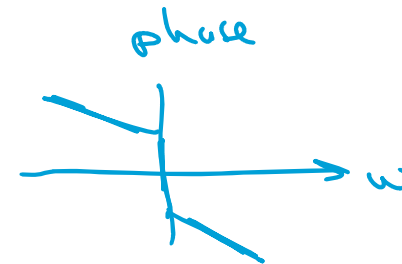
Magnitude

- $|e^{-j\omega n_0}| = 1$
- A pure delay does not change spectral magnitude



Phase

- Adds $-\omega n_0$ radians
- Linear phase slope encodes delay



Audio connection

An ideal sample delay preserves the spectrum's magnitude exactly; it only rotates phase versus frequency.

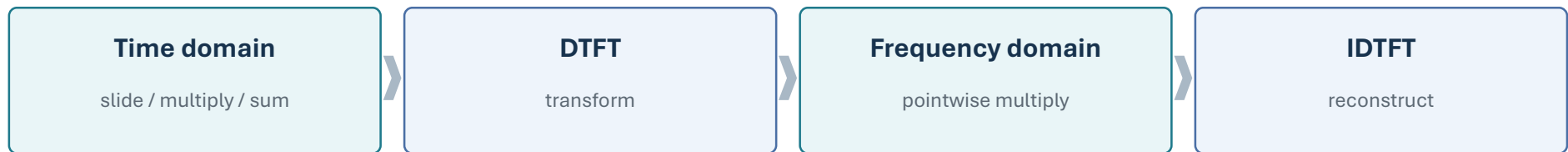
Convolution in time $\leftrightarrow$ multiplication in frequency

35-55 MIN

This is one of the most important DSP simplifications.

$$y[n] = x[n] * h[n]$$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$$



Proof: $y[n] = \sum_k x[k]h[n-k]$ Take DTFT $\sum_n \sum_k x[k]h[n-k]e^{-j\omega n}$

change
order of
summation

Computational echo

FFT-based convolution exploits the same multiplication idea numerically for long signals.

$$= X(e^{j\omega})H(e^{j\omega})$$

$$\sum_k x[k] \sum_n h[n-k] e^{-j\omega n} \xrightarrow{m \leftarrow n-k} \sum_k x[k] \sum_m h[m] e^{-j\omega(m+k)} = \sum_k x[k] e^{-j\omega k} \sum_m h[m] e^{-j\omega m} = X(e^{j\omega})H(e^{j\omega})$$

Parseval: energy is preserved across representations

35-55 MIN

The spectrum redistributes energy; it does not create it.

$$\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

Time-domain view

- Energy = sum of squared samples
- Natural for waveforms

Frequency-domain view

- Energy density across ω
- Natural for bands and features

Parseval's relation equates signal energy in time and freq domains

ML bridge

Many audio features—band energies, log spectra, mel features—summarize how signal energy is distributed across frequency.

Difference equations give $H(e^{j\omega})$ directly

35-55 MIN

Use the eigenfunction or DTFT property instead of solving every input separately.

$$\sum_{k=0}^N a_k y[n-k] = \sum_{m=0}^M b_m x[n-m] H(e^{j\omega}) = \frac{\sum_{m=0}^M b_m e^{-j\omega m}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$

$$x[n] = e^{j\omega n}$$

$$y[n-k] = H(e^{j\omega}) e^{j\omega(n-k)}$$

- Substitute $x[n] = e^{j\omega n}$, $y[n] = H(e^{j\omega}) e^{j\omega n}$.
- Factor out $e^{j\omega n}$.
- Solve the remaining algebraic equation for $H(e^{j\omega})$.

$$\sum_{k=0}^N a_k H(e^{j\omega}) e^{j\omega(n-k)} = \sum_{m=0}^M b_m e^{j\omega(n-m)}$$

$$H(e^{j\omega}) \sum_{k=0}^N a_k e^{-j\omega k} e^{j\omega n} = \sum_{m=0}^M b_m e^{j\omega n} e^{-j\omega m}$$

$$\text{Solve for } H(e^{j\omega}) = \frac{\sum_{m=0}^M b_m e^{-j\omega m}}{\sum_{k=0}^N a_k e^{-j\omega k}}$$

One-pole example: $y[n]=0.5y[n-1]+x[n]$

A simple recursive smoother with a strong low-frequency preference.

$$y[n] - 0.5y[n-1] = x[n]$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega}}$$

$$h[n] = \underline{(0.5)^n u[n]}$$

$$\Leftrightarrow H(e^{j\omega}) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}} \quad \text{if } |\frac{1}{2}| < 1$$

At $\omega=0$

- $H=1/(1-0.5)=2$
- DC is amplified

$$\omega = 0 : |H| = \frac{1}{1 - \frac{1}{2}} = 2$$

$$\omega = \pi : |H| = \frac{1}{1 + \frac{1}{2}} = \frac{2}{3} = 0.67$$

At $\omega=\pi$

- $H=1/(1+0.5)=2/3$
- Nyquist-frequency alternation is attenuated

$$\omega = \frac{\pi}{2} : |H| = \left| \frac{1}{1 - \frac{j}{2}} \right| = \frac{2}{\sqrt{5}} \approx 0.9$$

Think–Pair–Share: sketch $|H(e^{j\omega})|$ before computing

55–75 MIN

Reason from a few strategically chosen frequencies.

Given $y[n] = 0.5y[n-1] + x[n]$

- 1 minute: derive $H(e^{j\omega})$.
- 2 minutes: evaluate $|H|$ at $\omega=0, \pi/2, \pi$.
- 2 minutes: sketch a smooth 2π -periodic magnitude curve.
- Pair discussion: what audio behavior would you expect?

Prompt

What feature of the recursion causes “memory” of prior samples?

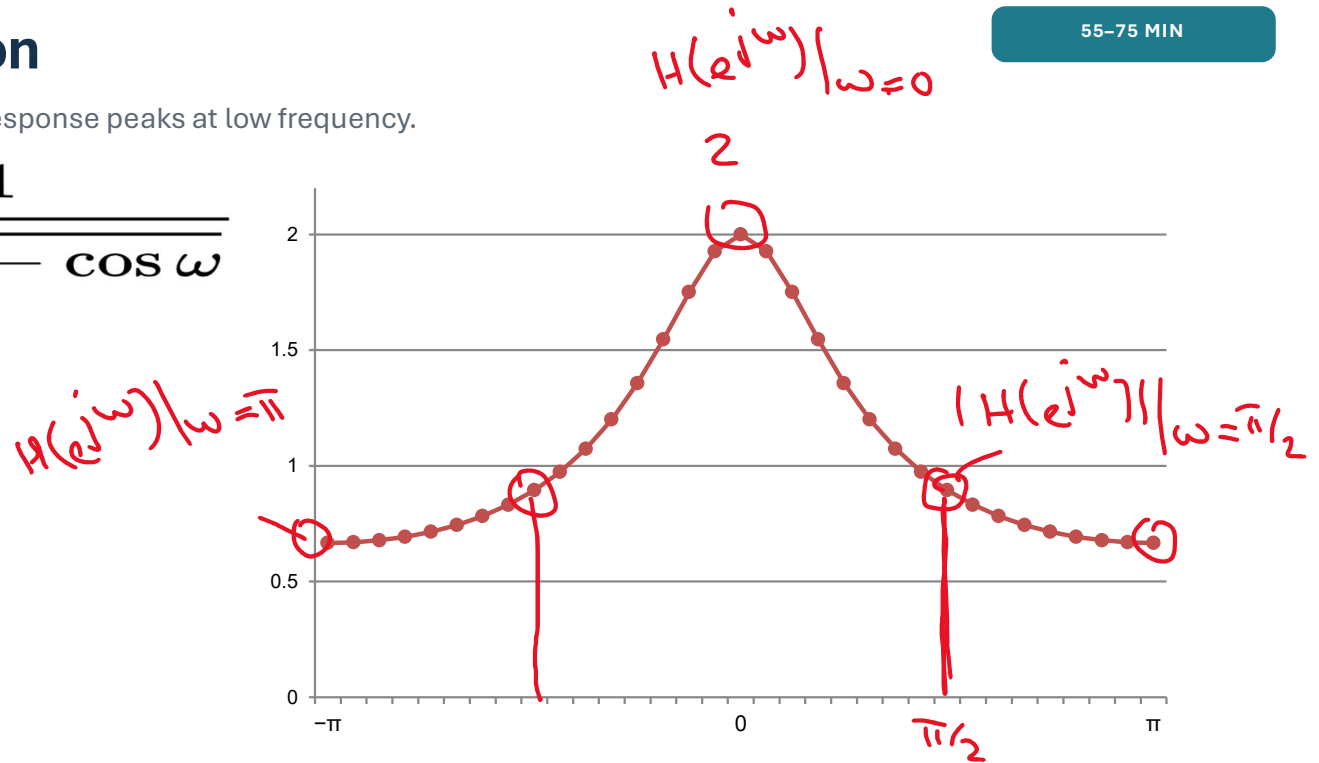
Graduate extension

Relate the coefficient 0.5 to a pole at $z=0.5$ and explain why moving it toward 1 increases low-frequency selectivity.

Think-Pair-Share solution

The denominator is smallest near $\omega=0$, so the response peaks at low frequency.

$$|H(e^{j\omega})| = \frac{1}{\sqrt{1.25 - \cos \omega}}$$



Three checkpoints

$\omega=0$: 2 • $\omega=\pi/2$: 0.894 • $\omega=\pi$: 0.667

Interpretation

Smooth low-pass tendency • recursive memory • stable because $|0.5| < 1$

Python demo: freqz and filtering

55–75 MIN

The numerical tools should confirm—not replace—the analysis.

SciPy frequency response

```
import numpy as np
from scipy import signal

b = [1.0]
a = [1.0, -0.5]
w, H = signal.freqz(b, a, worN=1024)
mag = np.abs(H)
```

Filtering

```
x = np.random.randn(48000)
y = signal.lfilter(b, a, x)
# Compare spectra / listen if desired
```

Notebook task

Change the recursion coefficient from 0.5 $\rightarrow$ 0.9 $\rightarrow$ 0.99. Predict the shape first; then verify with freqz.

DTFT vs DFT vs FFT

55-75 MIN

Three related terms that students often collapse into one.

DTFT

- Input: discrete-time sequence
- Frequency: continuous ω
- 2π -periodic spectrum
- Mathematical transform

DFT / FFT

- Input: finite N samples
- Frequency: N discrete bins
- DFT is the transform
- FFT is an algorithm for the DFT

$k=0 \dots N-1$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

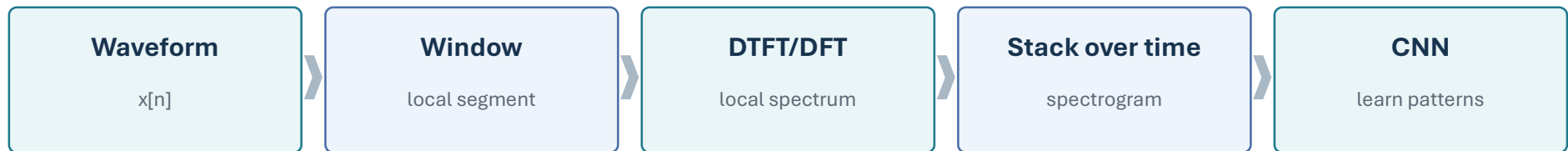
basis function
 $e^{-j2\pi kn/N}$

Practical bridge

In code, we sample the DTFT-like spectrum on a finite grid using the DFT/FFT. The grid and window matter.

ML bridge: spectrograms are structured frequency-domain features

The DTFT is the conceptual ancestor of the STFT.



- DSP explains what the axes and magnitudes mean.
- Windowing and sampling choices determine resolution and leakage.
- ML operates on features whose structure comes directly from signal-processing choices.

Concept check

CHECK

Reason from the denominator, not from memorized filter labels.

For $H(e^{j\omega}) = 1/(1 - 0.9e^{-j\omega})$, where is $|H|$ largest?

A. Near $\omega=0$

B. Near $\omega=\pi/2$

C. Near $\omega=\pi$

D. It is constant for all ω

Concept check

CHECK

Reason from the denominator, not from memorized filter labels.

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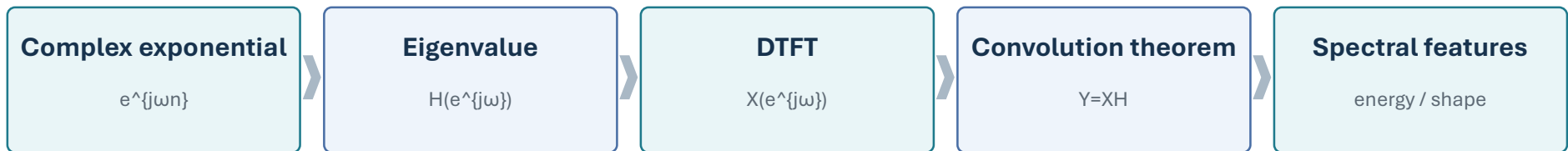
D. It is constant for all ω

Correct: at $\omega=0$ the terms 1 and $0.9e^{-j\omega}$ nearly cancel in the denominator, making the magnitude large.

Lecture 2 synthesis

WRAP-UP

The frequency domain turns LTI systems into multiplication.



- Be able to derive $H(e^{j\omega})$ from $h[n]$ or a difference equation.
- Be able to explain 2π periodicity and the relation $\omega=2\pi f/f_s$.
- Be able to reason qualitatively about a response before plotting it.

Week 2

Sampling will explain which analog frequencies map to the same discrete-time frequency—and why aliasing occurs.